

20	A cylindrical piece of wire has resistance R. It is stretched uniformly so that its length becomes three times longer, but its volume remains constant. What is its new resistance in terms of R?
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A	$\frac{R}{3}$	B	R	C	$3R$	D	$9R$
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Answer: D

Let original length = L , area = A .

Then original volume = $A \times L$

New length = $3L$

New area = $A' = \text{original volume} / \text{new length} = \frac{A \times L}{3L} = \frac{A}{3}$

Original resistance:

$$R = \frac{\rho \times L}{A}$$

New resistance:

$$R' = \rho \times \frac{3L}{\frac{A}{3}} = \rho \times (3L \times \frac{3}{A}) = 9 \times (\rho \times \frac{L}{A}) = 9R$$

21	A negative temperature coefficient thermistor, an LDR and a fixed resistor are connected in series
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